

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I K. Nehareka (192411124) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K. Nehareka

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Intelligent Agents In Smart Home Automation

Introduction :-

An Intelligent agent is a System that senses its Environment and it takes Suitable actions to achieve a goal. In a Smart home, Sensors, detect motion, temperature, time & Security Conditions and the agent Controls lights, fans, Ac & alarms.

1. Simple Reflex Agent :-

A Simple reflex agent acts only according to the Current Situation using Condition - action Rules.

Examples :-

- * If motion is detected $\rightarrow$ turn ON the light
- * If Smoke is detected $\rightarrow$ turn ON the alarm

2. Model Based Agent :-

- * It maintains an Internal State of the environment.
- * It uses both current & previous Information.

Ex :-

It remembers that a person is already present in a room & maintains the Status of lights & Temperature.

3. Goal Based Agent :-

A Goal based agent Selects actions to achieve a Specific Goal.

Ex :-

Its goal may be to maintain a Comfortable temperature While keeping the house Secure.

4. Utility Based Agent ÷

A utility based agent chooses the action that provides the best overall result by considering factors such as Comfort, energy consumption & security.

Ex ÷

It may use a fan instead of AC when the temperature is only slightly high to save energy.

Best Approach ÷

For a smart home, a hybrid approach using model based, goal based & utility based agents would be most effective.

A simple reflex component can handle immediate actions such as detecting motion (or) smoke. A model based component can maintain information about the current state of the house. A goal based component can work towards goals such as comfort & security. Finally, a utility based component can select the best action by balancing comfort, energy efficiency, cost & security.

Conclusion ÷

Therefore, a hybrid intelligent agent system is better than using only one type of agent. It can respond quickly to real time events, remember the environment, learn user preferences, & make decisions that provide comfort, energy efficiency & security.

Finding a path in an Unknown Maze

Such as
That
Computer
Introduction ÷

A robot placed in an unknown maze does not know the complete layout of the environment. It must explore different paths until it finds the exit.

Since there is no prior knowledge or heuristic information, uninformed search strategies can be used.

Two important uninformed search methods are :

- * Uniform Cost Search (UCS)
- * Iterative Deepening Search (IDS)

Uniform Cost Search [UCS] ÷

- * Uniform Cost Search expands the node with lowest path cost
- * It uses priority queue to select the cheapest path

Working ÷

- * Start from the initial position
- * Generate possible neighboring positions
- * Calculate the cost of reaching each position
- * Select the position with the lowest total path cost.
- * Continue until the exit is reached.

Ex ÷

If different paths have different movemental costs, UCS chooses the path with minimum total cost.

Advantages ÷

- * Complete If Step Costs are positive
- * optimal because it finds the least Cost Solution
- * Useful When different paths have different Costs.

Disadvantages ÷

- * Can use a large amount of memory
- * It takes log time if there are many paths available
- * It doesn't use direction of the Goal.

Iterative Deepening Search (IDS) ÷

Iterative Deepening Search Combines the advantages of depth first Search (DFS) & Breadth first Search (BFS).

- * It repeatedly performs depth limited DFS with increasing depth limits.

Ex ÷

Depth 0 → Search
 Depth 1 → Search
 Depth 2 → Search
 Depth 3 → Search
 ⋮
 exit

It Continues until the robot finds the exit.

Advantages ÷

- * Requires much less memory than BFS
- * Complete When the branching factor is finite.
- * finds the Shallowest Solution When all Step Costs are equal.

Advantages ÷

repeats the Same nodes Multiple times.

* Can Waste Computation

* Not optimal When different actions have different Costs

UCS vs IDS ÷

Feature	UCS	IDS
Selection	Lowest path cost	Increasing depth limit
Complete	Yes, with positive costs	yes
optimal	yes	yes only for equal steps costs
Memory	High	Low
Suitable for different costs	yes	No
Advantage	Minimum Cost path	Memory efficient

Conclusion ÷

Therefore, a normal maze where Each movement has the Same Costs, a model based & Goal based agent Combined With IDS is an effective choice because it uses less memory & can find the exit even when the Solution depth is unknown.